Certified Positivity: Formally Verified Horizon Thresholds for Truncated Weil Quadratic Forms in Lean

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Abstract

Weil's criterion reformulates the Riemann Hypothesis as the positivity of a quadratic form built from the primes, and a recent line of work (Connes–van Suijlekom; Connes–Consani–Moscovici) studies truncations of this form numerically, in floating point. We present a Lean 4/Mathlib formalization programme that treats the truncated setting with machine-checked certificates instead: a library of 137 machine-checked theorems spanning a structural skeleton (Cauchy interlacing, finite Osterwalder–Schrader reconstruction, the finite KMS identity, Sylvester's criterion, a Carathéodory–Toeplitz atomic-representation ladder, Lee–Yang and Hermite–Biehler steps) and a certified computation pipeline (kernel-pure rational enclosures of the prime atoms $\log p/\sqrt{p}$, Weyl-perturbation margin transfer, and positivity/non-positivity certificates for successive "horizon windows" of a discretized Weil-type form as each prime enters). The pipeline formally *decided* a pre-registered structural hypothesis — that the newest prime's atom binds each window's threshold — refuting it from the third window on, and certifying that from window four onward no single prime accounts for the threshold, anchored by a proof that $(\log p)/\sqrt{p}$ is globally maximized at $p = 7$. We further transcribe Suzuki's zeta screw function into Lean and certify that its genuine Gram kernel is positive definite on an explicit grid with an explicit margin — to our knowledge the first machine-checked instance of a positivity window for the true kernel, where the literature gives only an existence theorem — and that positivity persists across the first prime's entry. We describe the prover-in-the-loop workflow (specification, proof, hypothesis decision, error correction — including a case where the prover corrected a false numerical enclosure in the human specification), the design of certificate schemas that let exact rational verification compose with analytic error bounds, and lessons for formalizing frontier analytic number theory above the recently merged zeta/L-function stratum of Mathlib.

1 Introduction

The explicit formula of prime number theory is a pairing between the primes and the nontrivial zeros of the Riemann zeta function. Weil [15] recast the Riemann Hypothesis (RH) through this pairing: RH holds if and only if a certain quadratic form — assembled from the values of a test function at the points $\pm \log p^k$, weighted by $\Lambda(p^k)/p^{k/2}$, together with an archimedean term — is positive semidefinite on a suitable cone of test functions. Positivity of an explicit, concretely computable form is an unusual shape for a millennium problem to take, and it has recently become the object of a sustained computational programme: Connes and van Suijlekom [5] study a truncated Weil form whose parameter controls which primes enter, and prove that the associated ground state has its zeros on the critical line; Connes, Consani and Moscovici [4] construct self-adjoint operators from rank-one perturbations of the scaling action, with the same truncation structure; and independent implementations now compute these truncations at scale [2], tracking smallest eigenvalues and ground-state stability as the prime cutoff grows.

All of this computation is floating point. None of it is certified.

This paper reports a formalization programme that develops the truncated setting inside Lean 4 and Mathlib [14] with machine-checked certificates as the basic currency. The programme has two layers. The *structural layer* formalizes the finite-dimensional theorems that make truncated positivity analysis sound: Cauchy interlacing (shrinking a window can only raise the spectral floor), Sylvester's criterion (deciding positive definiteness by leading principal minors), a Weyl perturbation schema (certified floors survive certified errors), a finite Osterwalder–Schrader reconstruction and the finite KMS identity (the reflection-positivity and thermal structure that the Weil form's symmetry suggests), and a Carathéodory–Toeplitz ladder (finite-rank positive Toeplitz matrices are sums of exponential atoms) — the tool underlying self-adjointness arguments in the operator-theoretic approach [4]. The *computation layer* turns these schemas into certified numerics: kernel-pure rational enclosures of the constants that actually enter the truncated form — the prime atoms $(\log p)/\sqrt{p}$ for $p \leq 11$ — and then, window by window as each prime enters, formally verified thresholds for positivity of a discretized Weil-type form.

Working this way produced something floating-point experiments cannot: a *formally decided* structural hypothesis. At the second window the certified threshold coincided with the newest prime's atom, suggesting the hypothesis that the newest prime always binds. Pre-registered and tested at the third window, the hypothesis was *refuted by proof*: the certified threshold lies strictly above the newest atom's enclosure. At the fourth window the refutation was upgraded to a theorem of the discretization family: we prove that $(\log x)/\sqrt{x}$ is strictly decreasing on $[8, \infty)$, hence that the atom at $p = 7$ is the global maximum over all primes, and that the window-four threshold exceeds it — so from window four onward, *no single prime's atom can account for the threshold*; the binding is jointly produced. The certified threshold chain $0.6 < 0.789 < 0.919$ across windows 2–4 is, to our knowledge, the first formally verified quantitative data about truncations of a Weil-type form.

We make no claims about the Riemann Hypothesis. The certified windows concern a stated discretization family (Section 4); the bridge to the true kernel is built but awaits its constants (Section 5). The contribution is the corpus, the certificate architecture, and the workflow.

Contributions.

- A Lean 4 library of structural results around finite Weil-type positivity: interlacing `[D1:D1_a]`, finite Osterwalder–Schrader `[D2:D2_b_posdef]`, the finite KMS identity `[D3:D3_kms]`, Sylvester's criterion `[D6:D6_sylvester]`, and a Carathéodory–Toeplitz ladder `[F1:F1_rankone]`, `[F2:F2_c]`, `[F3R:F3R_unimodular]`.
- A certified computation pipeline: kernel-pure enclosures of $\log p/\sqrt{p}$ for $p \le 11$ `[G1:G1_log2]`, margin-transfer schemas `[D5:D5_b]`, `[K3:K3_main]`, and positivity/non-positivity certificates at successive horizon windows `[E4:E4_b_pos]`, `[E5:E5_cert_pos]`, `[G3:G3_cert_pos]`, `[G6:G6_cert_pos]`.
- Formally decided structure: refutation of the newest-prime-binds hypothesis `[G3:G3_newest_binds]`; the certified global atom maximum at $p = 7$ `[G5:G5_ord, G5_c_large]`; the joint-binding theorem `[G6:G6_c]`; and the certified threshold growth chain $0.6 < 0.789 < 0.919$ `[G6:G6_d]`.
- A prover-in-the-loop workflow for experimental mathematics, with certificate schemas that let exact rational proofs compose with analytic tails, and a faithfulness-audit protocol (Section 6).

The full Lean development accompanies the submission as an anonymized supplementary archive; every result cited in the text carries a `[file:declaration]` tag resolving into the archive's index.

2 Background: the Finite Objects

No analytic number theory is needed to state the objects this paper certifies; everything is finite-dimensional linear algebra plus rational enclosures of a handful of real constants.

2.1 The Weil form and its atoms

For a test function f and a truncation level N, the finite Weil-type functional is

$$W(f) \;=\; A(f) \;-\; \sum_{n=2}^{N} \frac{\Lambda(n)}{\sqrt{n}} \big(f(\log n) + f(-\log n)\big),$$

where Λ is the von Mangoldt function ($\Lambda(p^k) = \log p$, zero off prime powers) and A is an archimedean term. We formalize W as a Lean object and prove its linearity, its collapse on even test functions, and that its sum ranges effectively over prime powers `[GW1:GW1_a–d]`. The *atom* of a prime p is the leading weight $(\log p)/\sqrt{p}$ attached to the sites $\pm \log p$. Restricting attention to test data supported in a window $[-a, a]$ means only primes with $\log p \le a$ contribute: primes enter the form one at a time, at $\log 2, \log 3, \log 5, \ldots$ — we call these entry events *horizon crossings*. Positivity of the Weil form for all windows is Weil's criterion; positivity window-by-window is what we certify, for the discretization below.

2.2 The discretized window family

Fix the first k primes and place symmetrized sites at their logarithms. Our discretization family assigns a diagonal (archimedean) parameter κ and couples sites at distance d with the d-th atom, yielding a symmetric Toeplitz matrix; e.g. the window-4 form is the 5×5 matrix $U_5(\kappa, u, v, w, x)$ with first row (κ, u, v, w, x), instantiated at $u = (\log 2)/\sqrt{2}$, $v = (\log 3)/\sqrt{3}$, $w = (\log 5)/\sqrt{5}$, $x = (\log 7)/\sqrt{7}$. The *threshold* of a window is the critical κ separating positive definiteness from its failure. Parameterizing κ rather than fixing a quadrature makes the certified output — the threshold structure at each prime entry — independent of any particular archimedean discretization; certified κ values from a specific quadrature can be substituted later without touching the pipeline (Section 5).

2.3 The screw-function picture

The window family is a schematic shadow of a canonical object. Following the unification of the Weil-positivity literature through Krein's screw functions [8, 13], positivity of the Weil form is a statement that an associated even function g with $g(0) = 0$ has positive semidefinite *Gram kernel* $G_g(t, s) = g(t) + g(s) - g(t - s)$ on every finite grid. We formalize the screw-to-Gram map, prove that screw-type functions on a fixed grid form a convex cone `[K2:K2]`, and prove that the *free* case $g(t) = |t|$ has Gram kernel $2\min(t_i, t_j)$, which is positive definite on every increasing positive grid `[K1:K1_posdef]`. The true kernel is the free part plus a prime correction; Section 5 gives the certified margin-transfer schema that reduces positivity of the truth to entry-wise enclosures of the correction.

2.4 Relation to Mathlib's zeta stratum

Mathlib recently acquired the Riemann zeta function and Dirichlet L-functions, with the Euler product, functional equation, and a formal statement of RH [11], and a formalization of the Prime Number Theorem is under way [7]. That stratum provides the *objects*; the present development is a *quantitative certificate* layer above it, and touches the analytic stratum only through the von Mangoldt function

Table 1. Corpus map. Counts are headline theorems; supporting lemmas excluded. Axioms: ⋆ = `propext`, `Classical.choice`, `Quot.sound` only; † = additionally `Lean.ofReduceBool`/`trustCompiler` via one legacy enclosure file (Section 6).

Tier	Content	Thms	Axioms
A	Transvection/gauge skeleton	5	⋆
B	Hankel rank, recurrence, resolvent	4	⋆
C	Kolmogorov, Doob, entropic bridge	3	⋆
D	Interlacing, OS, KMS, Weyl, Sylvester	5	⋆
E	Enclosures; windows 1–2	5	†
F	Carathéodory ladder	3	⋆
G	Kernel-pure enclosures; windows 3–4	6	⋆/†
H	Flux, Lee–Yang, gcd positivity	6	⋆
I	Multiplicativity ⇔ product states	3	⋆
J	Box vs. Fejér windows, Hermite–Biehler	3	⋆
K,T	Screw kernel, margin transfer, tails	6	⋆
GW	Finite Weil functional	1	⋆

($\Lambda(n) \le \log n$, `[T1:T1_vonMangoldt_le_log]`) and kernel-pure enclosures of specific logarithms.

3 The Certified Corpus

The corpus was produced in four batches; Table 1 maps its tiers. This section walks the structural layer in the order the certification pipeline consumes it.

3.1 Horizon structure

Three theorems give the window family its shape. *Interlacing* `[D1:D1_a]`: deleting the last site of a window can only raise the spectral floor — so positivity restricts inward, and thresholds grow monotonically as horizons expand. *Finite Osterwalder–Schrader* `[D2:D2_b_posdef]`: a reflection-positive pairing across a window's midpoint descends to a genuine inner product on the quotient by its null space — the finite form of the reconstruction that turns reflection positivity into a Hilbert space. *Finite KMS* `[D3:D3_kms]`: for a Gibbs state $\omega(x) = \mathrm{Tr}(e^{-\beta H}x)/Z$ and the twist $\sigma(b) = e^{-\beta H} b e^{\beta H}$, we prove $\omega(a\,\sigma(b)) = \omega(b\,a)$ — thermal states are exactly twisted-trace states. The latter two results are not consumed by the computation pipeline; they formalize, at finite level, the structural reading of *why* a reflection-symmetric form built from a one-parameter flow could be expected to be positive, and they shaped the library's design.

3.2 Certification schemas

The engine has two moving parts. *Sylvester's criterion* `[D6:D6_sylvester]` — proved in full, including the Schur-complement induction and a bridge to Mathlib's `Matrix.PosDef` — turns "is this explicit rational matrix positive definite" into a finite determinant computation. The *Weyl schema* `[D5:D5_b]` — $\lambda_{\min}(A + E) \ge \lambda_{\min}(A) - \varepsilon$ when $|x^{\mathrm{T}}Ex| \le \varepsilon$ on unit vectors — transfers a certified floor across a certified error, and its entrywise form `[K3:K3_main]` ($|R_{ij}| \le \delta$ implies $\varepsilon \le n\delta$) lets rational matrices with enclosure-bounded deviations stand in for real ones. Together they make exact rational proof compose with analytic bounds: prove the rational approximant positive definite with margin, bound the deviation, conclude for the truth.

3.3 Enclosures

The constants entering windows 1–4 are $\sqrt{p}$ and $\log p$ for $p \in \{2, 3, 5, 7\}$ (and 11 for the maximality theorem), and the atoms $(\log p)/\sqrt{p}$. All are enclosed in rationals at 10^{-4} width, e.g. $0.4900 < (\log 2)/\sqrt{2} < 0.4902$ and $0.6342 < (\log 3)/\sqrt{3} < 0.6344$ `[E3:E3_atom2]`. An early batch proved the logarithm bounds by `native_decide`; a later batch re-derived them kernel-purely `[G1:G1_log2]` from Mathlib's digit lemmas for $\log 2$ and truncated alternating series for $\log(1 + x)$. One further `native_decide` use remains: an integer power comparison in the maximality tail bound `[G4:]`, inherited by the atom-maximality and joint-binding theorems (Section 6 records the full axiom status).

3.4 The Carathéodory–Toeplitz ladder

The structural crown is the finite atomic-representation theorem: a positive semidefinite Hermitian Toeplitz matrix of rank r is a sum of r rank-one exponential atoms $c_m = \sum_s \rho_s e^{im\theta_s}$, $\rho_s > 0$ [3]. This is simultaneously the tool underlying self-adjointness in the operator-theoretic programme [4] and, in moment language, the statement that finite-rank positivity forces an atomic spectral measure. Its proof is absent from Mathlib (no Fejér–Riesz theory), so we built it as a ladder: the rank-one case in full `[F1:F1_rankone]`; Vandermonde uniqueness, reality, and strict positivity of the coefficients given distinct unimodular nodes `[F2:F2_a]`–`[F2:F2_c]`; and unimodularity of the annihilator's roots `[F3R:F3R_unimodular]`, proved via the Szegő route — the Toeplitz shift identity makes multiplication by z an isometry of the finite form, descending to a unitary on the quotient whose characteristic polynomial is the annihilator. The sole remaining gap in the entire corpus is the *simplicity* of these roots, whose standard proof needs diagonalizability of a unitary on a finite-dimensional inner-product space — currently absent from Mathlib, and hence an upstreaming target rather than a research gap (Section 8).

3.5 Supporting tiers

Four smaller tiers certify facts that the analytic literature uses constantly and that fit naturally in the library. *Sharp windows break positivity*: the box Toeplitz matrix with first row $(1, 1, 0)$ is not PSD, while every autocorrelation (Fejér-type) Toeplitz matrix is PSD as an adjoint square `[J1:J1_a]`, `[J1:J1_b]` — test windows must be mollified to stay in the positive cone. *Hermite–Biehler* `[J2:J2_hermite_biehler]`: a polynomial with

Table 2. Certified window thresholds. κ_0: proved positive definite for all atom values within their enclosures; κ_1: proved not positive definite likewise. The window-1 threshold is exact.

Window	Primes	Size	Not PD at	PD at
1	2	2×2	$\kappa^* = (\log 2)/\sqrt{2}$ (exact)	
2	2,3	3×3	0.600	0.635
3	2,3,5	4×4	0.789	0.790
4	2,3,5,7	5×5	0.919	0.921

all roots in the lower half-plane satisfies $|E(\bar{z})| < |E(z)|$ above the axis — the defining inequality of the de Branges structure in which the canonical-system reading of the Weil form lives. *Lee–Yang steps* `[H3a:H3a_i]`, `[H3b:H3b_asano]`: the two-spin base case and the Asano contraction — the mechanism by which coupling positivity forces zeros onto the critical locus [10]. *One-prime purity* `[J3:J3_half]`: the zeros of $1 - c\,e^{-sL}$ all have real part $(\log c)/L$, equal to $\frac{1}{2}$ iff $c = e^{L/2}$ — at the level of a single Euler factor, critical-line alignment is exactly the purity of the atom's weight. These are one-page certificates individually; collectively they pin the positivity grammar the main pipeline instantiates.

4 The Horizon March

Table 2 is the paper's experimental heart: the certified threshold at each horizon crossing. Each row is a pair of theorems — positive definiteness for every atom value inside the certified enclosures at κ_0, and its failure at κ_1 — so the true threshold of the discretized family is formally pinned inside $(\kappa_1, \kappa_0]$.

4.1 Window 1: an exact threshold

The first window is a competition between the archimedean diagonal and the single prime 2: $W(\kappa, w)$ is PSD iff $\kappa \geq |w|$ `[E4:E4_a]`, so the threshold is exactly the atom $(\log 2)/\sqrt{2}$, certified against its enclosure `[E4:E4_b_pos]`. Entry of the first prime prices the archimedean term precisely.

4.2 Window 2 and a hypothesis

With primes 2, 3 inside, the certified thresholds are $\kappa_1 = 0.6$, $\kappa_0 = 0.635$ `[E5:E5_cert_pos]`; a representative certificate:

```
theorem E5_cert_pos :
  IsPDq (V 0.635 (Real.log 2 / Real.sqrt 2)
             (Real.log 3 / Real.sqrt 3))
```

The interval $(0.6, 0.635]$ contains the newest atom $(\log 3)/\sqrt{3} \approx 0.6343$, and inspection of the Sylvester factors shows the constraint essentially *is* that atom: the newest prime binds, the older prime's coupling is slack. This suggested a structural hypothesis — *the newest prime's atom binds every window's threshold* — which we pre-registered as a formal test for the next window.

4.3 Window 3: the hypothesis is refuted by proof

The window-3 certificates give $\kappa_1 = 0.789$, $\kappa_0 = 0.790$ `[G3:G3_cert_pos]`, and the decision theorem `[G3:G3_newest_binds]` proves $0.7198 \leq \kappa_1$: the certified threshold lies strictly *above* the newest atom's enclosure $(0.7197, 0.7198)$ for $(\log 5)/\sqrt{5}$. The hypothesis is false from window 3 on: the primes' pressures on the floor combine. A negative verdict, decided by proof rather than by reading a plot, is exactly the output the pipeline was designed to produce.

4.4 The dichotomy anchor

Whether joint binding persists could in principle need checking forever — unless single-atom pressure has a global ceiling. It does: $(\log x)/\sqrt{x}$ is strictly decreasing on $[8, \infty)$ `[G5:G5_ord, G5_c_large]`, and combining with the certified orderings of the first four atoms and the enclosure $(\log 11)/\sqrt{11} < (\log 7)/\sqrt{7}$ `[G4:G4]`, the atom at $p = 7$ is the *global maximum* over all primes, certified: every prime $p \neq 7$ has $(\log p)/\sqrt{p} < (\log 7)/\sqrt{7} < 0.7356$.

4.5 Window 4: joint binding becomes a theorem

The window-4 certificates give $\kappa_1 = 0.919$, $\kappa_0 = 0.921$ `[G6:G6_cert_pos]`, proved via a centrosymmetric factorization of the 5×5 Toeplitz determinant. Since $0.919 > 0.7356$ exceeds the certified global atom maximum, *no single prime's atom can account for the threshold at window 4 or beyond* `[G6:G6_c]`: within this discretization family, joint binding is a theorem, not an observation. The certified growth chain $0.6 < 0.789 < 0.919$ `[G6:G6_d]` is the formal shadow of the smallest-eigenvalue curves computed in floating point by the numerical programme [2].

4.6 Scope

These are certificates about the stated discretization family. Whether joint binding reflects the true Weil kernel or an artifact of distance-indexed coupling overlap is precisely the question the next section's machinery is built to settle.

5 Toward the True Kernel

The bridge from the schematic family to the genuine object is complete except for its constants. The screw-function picture (Section 2) writes the true kernel as free part plus prime correction. On the free side, the min-kernel is positive definite on every increasing positive grid `[K1:K1_posdef]`, with a certified positive floor `[K3:K3_main]`. On the correction side, the tail toolkit bounds every Λ-weighted tail that a truncation discards: $\Lambda(n) \leq \log n$ `[T1:T1_vonMangoldt_le_log]`, $\sum_{n>N} n^{-3/2} \leq 2/\sqrt{N}$ `[T2:T2_power_tail]`, and $\sum_{n>N} (\log n)\, n^{-3/2} \leq (2\log N{+}4)/\sqrt{N}$ `[T3:T3_log_tail]`. The margin-transfer schema then reduces certified positivity of the true Gram matrix on a prime-log grid to entrywise rational enclosures of finitely many kernel values beating the free part's certified floor

— the same Sylvester-plus-Weyl pattern that ran the horizon march, pointed at the real object. This transcription has now been carried out. We define the genuine screw function Ψ in Lean — archimedean part (with the constants $A = -\gamma - \pi/2 - 3\log 2$, $C = \zeta(2, \frac{1}{4})$, and the Hurwitz–Lerch series L) minus the von Mangoldt prime sum — with its Gram kernel $G(t, u) = \Psi(t) + \Psi(u) - \Psi(t - u)$ `[V5_1:G]`, and certify the needed constants ($\gamma \in (0.5772, 0.5773)$, $C \in (17.1972, 17.1974)$, $A \in (-4.2276, -4.2273)$) `[V5_2:A_bounds]`, `[V5_3:C_bounds]`. Two results follow on the genuine kernel. First, the 3×3 Gram on the prime-free grid $(0.2, 0.4, 0.6)$ is positive definite with certified least eigenvalue ≥ 0.005 `[V5_5:true_kernel_grid_posdef]`, `[V5_5:true_kernel_grid_margin]` — to our knowledge the first explicit machine-checked instance of Suzuki's Theorem 4.2, which establishes only that some such window exists. Second, the 2×2 Gram on $(0.4, 0.9)$ stays positive definite *across the first prime's entry* (the $(\log 2)/\sqrt{2}$ ramp switches on at $\log 2 \approx 0.693$ between the grid points) `[V5_6:true_kernel_first_prime_posdef]`. Finally, the finite Weil functional `[GW1:GW1_a-d]` and Ψ are certified to agree through triangular test data `[V5_7:weil_triangle_prime_side]`, welding the paper's two formal objects into one.

6 Prover-in-the-Loop Methodology

The corpus was produced by an iterated protocol between a specification side (the author working with an AI assistant, Claude by Anthropic) and an automated Lean 4 prover, in four batches totalling 137 theorems plus 33 supporting lemmas (per-batch headline yields 12/12, 10/11, 17/18, and 20/21). The protocol, and two incidents that stress-tested it, are a contribution in their own right.

6.1 The protocol

Each batch is a set of independent, self-contained natural-language specifications: finite-dimensional statements with explicit definitions, suggested proof routes, and declared acceptable partial credit. The prover returns Lean sources; an AI-assisted audit then (i) diffs every formal statement against its specification, recording strengthenings and weakenings in docstrings; (ii) censuses `sorry`s and confirms each is a declared partial-credit site; (iii) audits axioms per theorem; and (iv) propagates verdicts to the next batch's design. Faithfulness auditing is the load-bearing step: a prover can prove a subtly weaker statement, and only statement-level diffing catches it. In four batches the audit found several benign strengthenings (unused hypotheses dropped, results generalized from finite-dimensional to arbitrary vector spaces) and honest hypothesis additions (a nonempty-state-space assumption without which one equivalence is genuinely false at $n = 0$) — and zero unfaithful proofs.

For calibration, the corpus stratifies honestly by difficulty: roughly half the declarations are deliberately routine pipeline components (rational enclosures, one-line algebraic identities) whose value is compositional; some forty are standard-technique formalizations; and the load-bearing items — the unimodularity theorem via the Szegő route `[F3R:F3R_unimodular]`, Sylvester's criterion with its Schur-complement induction `[D6:D6_sylvester]`, the Kronecker recurrence `[B2:B2]`, the entropic-bridge development `[C3:C3]`, the Osterwalder–Schrader reconstruction `[D2:D2_b_posdef]`, and the centrosymmetric factorization behind window 4 `[G6:G6_cert_pos]` — carry the formalization weight. The library's contribution is architectural: how exact rational proof, spectral margin transfer, and analytic tails compose into decisions, not the per-item difficulty of its components.

6.2 The prover corrected the specification

One batch specified the enclosure $0.6343 < (\log 3)/\sqrt{3}$. This is false (the value is ≈ 0.634284). The prover detected the impossibility, substituted the correct enclosure $0.6342 < (\log 3)/\sqrt{3} < 0.6344$, adjusted the dependent window ranges, re-verified the downstream certificates unchanged, and documented the repair. Error flowed in the direction opposite to the usual worry: in this workflow the specification specification, not the machine proof, was the failure surface — across all four batches, every detected error originated in the specifications.

6.3 A hypothesis decided by proof

The newest-prime-binds hypothesis (Section 4) was formulated after window 2, pre-registered as a formal test in the window-3 specification, refuted by the resulting theorem, and converted at window 4 into a positive theorem (joint binding) with the maximality anchor. The sequence — observe, pre-register, decide by proof, generalize — is the experimental method with the ambiguity removed from the deciding step.

6.4 Axiom hygiene

All structural theorems use only `propext`, `Classical.choice`, `Quot.sound`. One early enclosure file proved logarithm bounds by `native_decide`, adding `Lean.ofReduceBool`/`trustCompiler` to its dependents; a later batch re-derived the bounds kernel-purely, and new certificates use the pure versions, with one exception: a single `native_decide` integer power comparison in the maximality tail bound (G4), which — together with the legacy E2 inheritance via G4's import — the atom-maximality and joint-binding theorems (G5/G6) inherit. All other structural theorems, including every true-kernel certificate of Section 5, remain on the three-axiom baseline. The status is recorded per theorem in the artifact's axiom audit (Appendix A).

6.5 AI disclosure

The Lean proofs were produced by an automated theorem prover from specifications developed by the author with AI assistance; all formal statements were audited against those specifications (AI-assisted) against their specifications as described above; the prose of this paper was likewise a product of that collaboration. The kernel checks what matters: every claimed theorem is machine-verified, and the per-theorem provenance table is Appendix B. This work complies with the ACM Policy on Authorship concerning the use of AI: the specifications, certificate-schema design, experimental design (pre-registration and hypothesis decisions), audit protocol, and library architecture constitute the authors' contribution; the prover is employed as an automated tool, and the authors accept responsibility for all material.

7 Related Work

Formalized analytic number theory. Mathlib's zeta/L-function stratum [11] provides the objects (Euler product, functional equation, the formal RH statement); the PNT project [7] is building the asymptotic stratum. The present corpus is a quantitative certificate layer above both, and overlaps neither in results. A recent independent Lean repository [12] formalizes Ihara zeta functions, the Bass formula, and *conditional* spectral reductions (self-adjointness of a suitable operator implies RH-type statements) together with abstract positivity classes. That work is logical-reduction-shaped; ours is certificate-shaped (verified numerical thresholds for explicit truncations); the two are complementary and disjoint.

Numerical truncated-Weil programme. The truncated Weil form and its operator realizations are developed in [4, 5], with independent large-scale implementations and a finite Guinand–Weil dictionary in [1, 2], all in floating point. Our thresholds are certified but (so far) schematic; theirs are uncertified but attached to the true form; K4 (Section 5) is the designed meeting point. The screw-function frame that our K-tier shadows at finite level is [8, 13].

Certified numerics in proof assistants. The enclosure-plus-schema pattern belongs to the tradition of verified numerical certificates in formal proof, of which the Kepler conjecture formalization is the landmark [6]; the specific composition here — Sylvester decisions on rational matrices transferred to real ones by Weyl margins against enclosure-bounded corrections — is tailored to positivity thresholds and, we believe, reusable wherever spectral floors meet analytic error terms. We note prior formal positive-definiteness infrastructure: a Cholesky/Sylvester characterization formalized in ACL2 [9], and Mathlib's Gershgorin circle theorem with strict-diagonal-dominance nonsingularity; our Sylvester development and the quantitative diagonal-dominance Rayleigh-floor lemma used downstream are, to our knowledge, not currently in Mathlib in the forms we require, though we make no priority claim beyond that.

8 Future Directions

Four concrete steps, in order. (1) *The true kernel*: transcribe the zeta screw function's finite sums into the enclosure format and run the existing pipeline — the first certified positivity windows of the genuine Weil Gram matrix, and the certified answer to whether joint binding is real or artifact. (2) *The finite Guinand–Weil dictionary*: formalize the bridge from finite windows to the explicit formula proper [1] — on paper but not in Lean, and the natural next stratum above [11]. (3) *Mathlib upstreaming*: the corpus's sole gap is diagonalizability of unitaries on finite-dimensional inner-product spaces; contributing it (with Fejér–Riesz and the completed Carathéodory theorem) closes our last `sorry` in the library where it belongs. (4) *Certified spectral instruments*: extend the march with certified effective-rank and realization-degree readouts of truncated kernels, alongside the thresholds.

Beyond these, one programmatic remark. The structural tier was not chosen at random: reflection positivity (D2) and the KMS identity (D3) are the finite faces of the thermal reading of the Weil form's symmetry, and the horizon march is the finite face of its criterion. The library is shaped so that if the analytic programme identifies the canonical object behind the truncated forms, the certificate layer is already pointed at it. No claims attach to this remark; it explains the library's architecture.

Acknowledgments

The Lean 4 proofs were produced by an automated theorem prover (Aristotle, Harmonic) from specifications the author developed in collaboration with an AI assistant (Claude, Anthropic); the statement-level audit of prover output against those specifications was likewise AI-assisted, as was the prose. The author set the problems, made the mathematical and editorial decisions, and takes responsibility for all claims. No human has reviewed the proof terms line by line; the trust model is Lean's kernel, with per-theorem axiom status recorded in the artifact; the consolidated build of the full development has been independently re-run on the pinned toolchain (exit code 0, axiom lists confirmed by `#print axioms`), with the recipe in the artifact. See the artifact's provenance table for per-batch detail.

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A Corpus Index and Axiom Audit

The supplementary archive contains the complete Lean development (four batch directories under a single project, building against a pinned Mathlib). Its `README` carries the full theorem index — for every headline theorem: file, declaration name, one-line statement, specification-diff notes, and axiom census — of which Table 1 is the summary. The three `sorry` sites (`D4`, `F3`, `F3R_simple`) share the single mathematical root documented in Section 3.4; each carries a docstring stating exactly what is assumed and why.

B AI-Assistance Transparency Table

Per batch: specification side (author + AI assistant), prover (automated Lean 4 theorem prover), audit steps performed (statement diff, sorry census, axiom check), corrections in either direction (one specification error corrected by the prover; zero unfaithful proofs detected), and turnaround. The full table accompanies the artifact.